

Measures of variability of discrete random distributions

Variance

- Important feature of probability distributions as it provides information about the spread of the distribution with respect to the **mean**.
- Large variance → it implies that the possible values are spread (or deviate) quite a distance from the mean.
Small variance → implies that the possible values are close to the mean.
- Variance is also called a **measure of spread or dispersion**.
- The variance is written as **Var(X)** and denoted by the symbol σ^2 (sigma squared). It is defined as the **expected value (or average) of the squares of the spreads (deviations) from the mean**.

Cont.

- The rule for variance is given by: $\text{Var}(X) = E(X - \mu)^2 = \sum (X - \mu)^2 \Pr(X = x)$.
- performing the calculation is quite a lengthy process.
- Alternative rule

$$\begin{aligned}\text{Var}(X) &= E(X - \mu)^2 \\ &= E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - E(2\mu X) + E(\mu^2) \\ &= E(X^2) - 2\mu E(X) + E(\mu^2) \\ &= E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2 \\ &= E(X^2) - [E(X)]^2\end{aligned}$$

since $E(X) = \mu$ (the mean)

Example 15

Find the expected value and variance of the following probability distribution table.

x	1	2	3	4	5
$\Pr(X = x)$	0.15	0.12	0.24	0.37	0.12

Find the variance of $(2X+1)$.

The variance of a linear function can also be calculated by the following rule:
 $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

Standard Deviation

- Another important measure of spread is the standard deviation.
- SD(X) or denoted by the symbol σ (sigma).
- The standard deviation is the **positive square root of the variance**.
- It is defined by the rule:

$$\begin{aligned}\text{SD}(X) &= \sqrt{\text{Var}(X)} \\ &= \sqrt{\sigma^2} \\ &= \sigma.\end{aligned}$$

- Variation and standard deviation are used extensively in many real-life applications involving statistics. **Analysis of data would be useless without any information about the spread of the data.**

Interpreting the standard deviation

- A characteristic of many distributions is that approximately 95% of the spread or distribution lies between 2 standard deviations of the mean; that is,
 - $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$.
- It is important to note that when calculating the $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$ for a specific distribution an exact value of 0.95 will not always be achieved, but should be close to it.

For many random variables, approximately 95% of the spread of the population lies between 2 standard deviations of the mean, that is,

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95.$$

Example 16

A probability distribution is shown below.

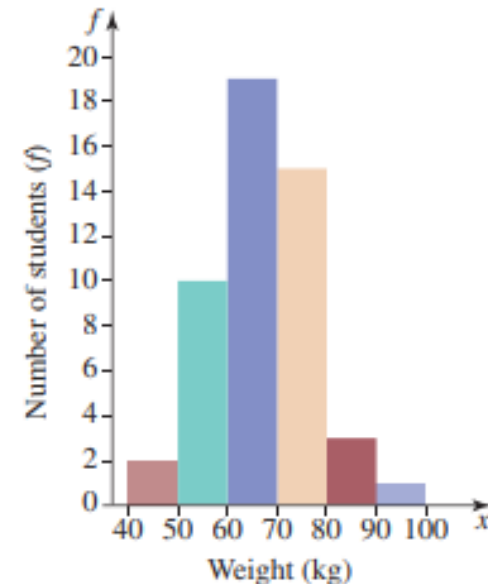
Given that $\mu = 3.38$ and $\sigma = 0.946$, calculate $\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

Continuous Random Variables

Introduction

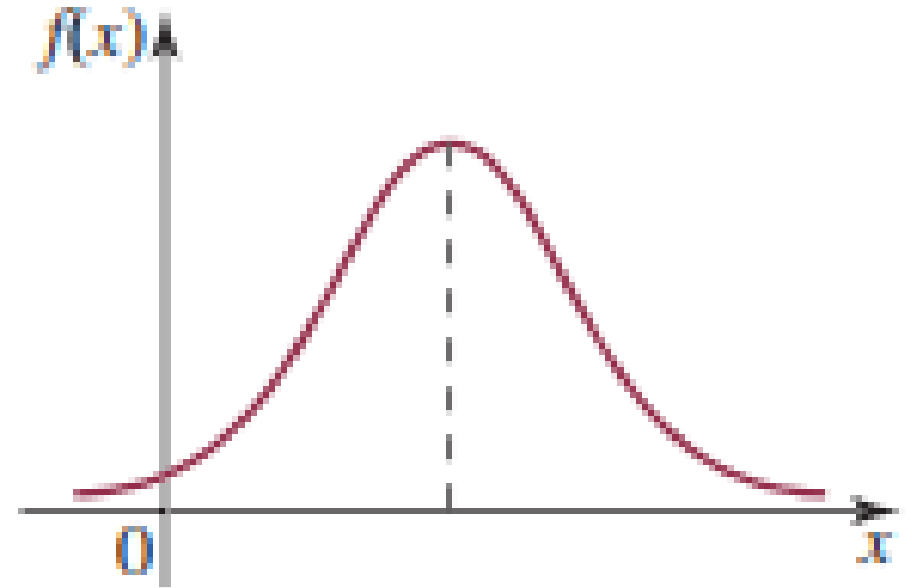
- Continuous random variables represent quantities that can be measured and thus may assume any value in a given range.
- They include such variables as time, height and weight.
- The weights of 50 Year 12 students are displayed in the table below, alongside a histogram.

Weight (kg) x	Frequencies f
40– < 50	2
50– < 60	10
60– < 70	19
70– < 80	15
80– < 90	3
90– < 100	1
Total = 50	



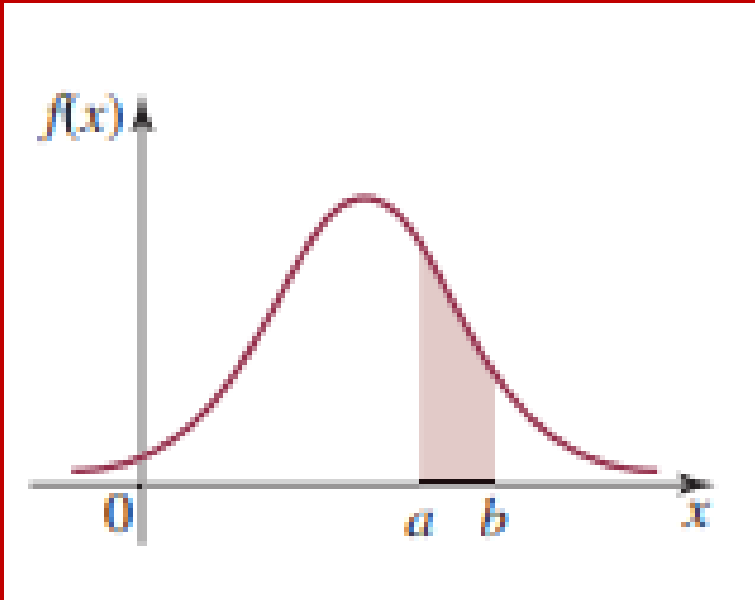
Cont.

- The frequency of individual weights cannot be determined because the weights have been grouped into class intervals. This limits the information we are able to extract from the histogram.
- The probability of a continuous random variable assuming an exact value is zero. In order to determine the probability of continuous random variables, a new method must be employed.
- If we were to increase the size of the sample of students' weights and make the class intervals very small, then the histogram would become a smooth frequency curve as shown here.



Probability density function

- A special scaled version of a smooth frequency curve is the probability density function or pdf. The scale is such that the probability of the random variable lying between certain values is given by the area between the pdf and the horizontal axis.



- The probability of X , a continuous variable, falling between $x = a$ and $x = b$ is represented by the shaded area between a probability density function $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$.
- The area is determined by integrating $f(x)$ from $x = a$ to $x = b$.

$$\Pr(a < x < b) = \int_a^b f(x) dx.$$

If X is a continuous random variable, then the probability of it falling between certain values is given by the area under a frequency curve, known as a probability density function or pdf.

Properties of a probability density function

1. $f(x) \geq 0$ for all x , (that is, we never have negative probabilities) and
2. $\int_{-\infty}^{\infty} f(x) dx = 1$ (that is, the sum of all probabilities is equal to one).

Remember:

- The domain of a pdf is usually $\mathbb{R}$, that is, the variable is continuous and can assume any real value (at least in theory).
- If we need our function to be non-zero over a particular interval only, say, over (a, b) , then we need to specify that the function is equal to zero everywhere else. For example,

$$f(x) = \begin{cases} \frac{2}{3}x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

- If a pdf is only non-zero over a particular interval (a, b) , the area under the curve between $x = a$ and $x = b$ must be equal to one. That is,

$$\int_a^b f(x)dx = 1.$$

Example 17

Sketch the graph of each of the following and state whether each function may be a probability density function (pdf).

a $f(x) = \begin{cases} 4x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$

b $f(x) = \begin{cases} 10x - 4, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$

c $f(x) = \begin{cases} 0.5, & -1 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$

Example 18

Find the value of a so that the given function may be a probability density function (pdf).

$$f: [2, 4] \rightarrow \mathbb{R}, f(x) = a(12 - 3x)$$

Using a probability density function to find probabilities of continuous random variables

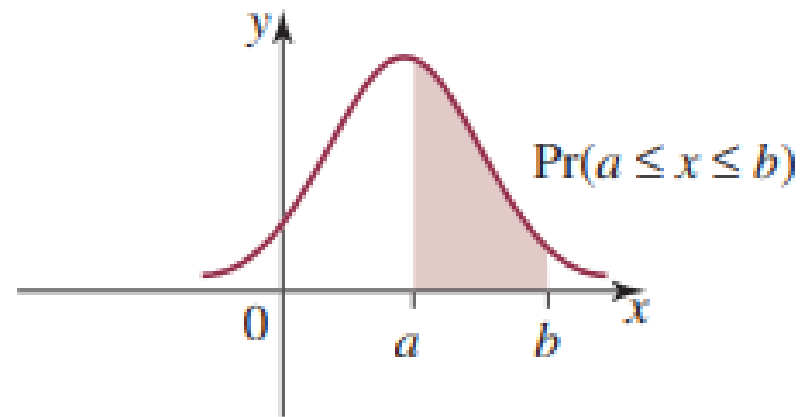
- If X is a continuous random variable, then the probability of it falling between $x = a$ and $x = b$ is given by the area bounded by the probability density function, the x -axis and the lines $x = a$ and $x = b$. This area can be easily calculated by evaluating the definite integral between a and b . That is,

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx.$$

- and

$$\int_a^a f(x) dx = 0, \text{ so } \Pr(X = a) = 0.$$

$$\therefore \Pr(X > a) = \Pr(X \geq a)$$



Example 19

If X is a continuous random variable with the probability density function defined as

$$f(x) = \begin{cases} \frac{1}{2}x - 4, & 8 \leq x \leq 10 \\ 0, & \text{elsewhere} \end{cases}$$

find:

a $\Pr(X > 9)$ **b** $\Pr(8.5 < X < 9)$ **c** $\Pr(X > 9 \mid X > 8.5)$.

Remember:

- In many instances we need to integrate over intervals involving infinity. In such cases, the limits are evaluated as shown at right.

Interval	Limit
$(-\infty, \infty)$	$\lim_{k \rightarrow \infty} \int_{-k}^k f(x) dx$
$(-\infty, a)$	$\lim_{k \rightarrow -\infty} \int_k^a f(x) dx$
(a, ∞)	$\lim_{k \rightarrow \infty} \int_a^k f(x) dx$

Quantiles and percentiles

- Quantiles and percentiles both define the value of the random variable, X , below which a given proportion of the distribution falls. A quantile is a probability value expressed as a decimal, while a percentile is a probability value expressed as a percentage.
- Ex:
 - if $\Pr(X < k) = 0.6$, then k is the 0.6 quantile, or the 60th percentile.
 - if we want to find the 60th percentile of a continuous random variable, X , with a probability density function given by $f(x)$ and $a \leq X \leq b$, solve for k the equation of the form

$$\int_a^k f(x)dx = 0.6.$$

Example 20

If X is a continuous random variable with a probability density function given by

$$f(x) = \begin{cases} 4 - 2x, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}, \text{ find the value of } a \text{ such that } \Pr(X \leq a) = 0.75, \text{ where } 1 \leq a \leq 2.$$

Measures of central tendency and spread

Mean

- If X is a continuous random variable with a probability distribution function $f(x)$ over the domain R , the mean of X can be found using the formula

$$\mu = E(x) = \int_{-\infty}^{\infty} xf(x)dx.$$

- If $f(x)$ has a certain rule for $a \leq X \leq b$ and is zero elsewhere, then the mean can be obtained by using the formula

$$\mu = E(x) = \int_a^b xf(x)dx.$$

- In general,

$$E[g(x)] = \int_a^b g(x)f(x)dx.$$

Median and Mode

- The median value, m , of the continuous random variable, X , is a value such that

$$\Pr(X \leq m) = \frac{1}{2}.$$

- To find the median of X we need to solve for m the equation

$$\int_a^m f(x)dx = \frac{1}{2}.$$

- The mode is a value of X for which $f(x)$ has its maximum.

Variance

- If X is a continuous random variable with the probability distribution function $f(x)$ over the domain R , the variance of X can be found using the formula

$$\text{Var}(X) = E(x - \mu)^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx.$$

- If $f(x)$ has a certain rule for $a \leq X \leq b$ and is zero elsewhere, then the variance can be obtained by using the formula

$$\text{Var}(X) = E(x - \mu)^2 = \int_a^b (x - \mu)^2 f(x) dx.$$

- The alternative formula for variance is given by $\text{Var}(X) = E(X^2) - \mu^2$. This can also be expressed as

$$\text{Var}(X) = \int_a^b (x)^2 f(x) dx - \left(\int_a^b x f(x) dx \right)^2.$$

Standard Deviation and IQR

- The standard deviation of X can be found by taking the positive square root of the variance of X . That is,
$$SD(X) = \sqrt{VarX}$$
- The interquartile range (IQR) is the middle 50% of the distribution.
IQR = 75th percentile – 25th percentile.
= $Q3 - Q1$

Example 21

Find the mean, median and mode for the following probability density function.

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the variance and standard deviation for the following probability density function.

$$f(x) = \begin{cases} x - \frac{1}{2}, & 1 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}.$$